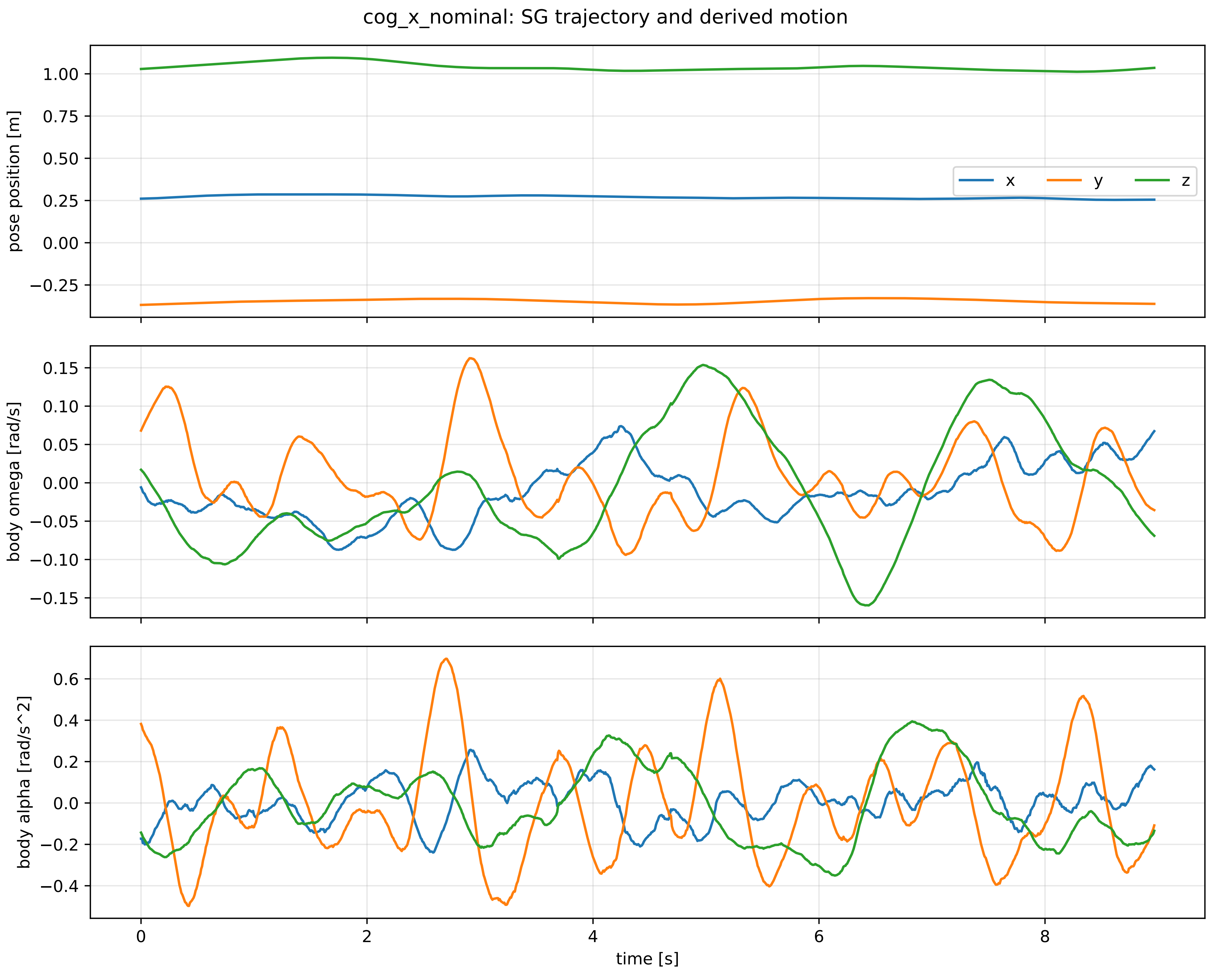
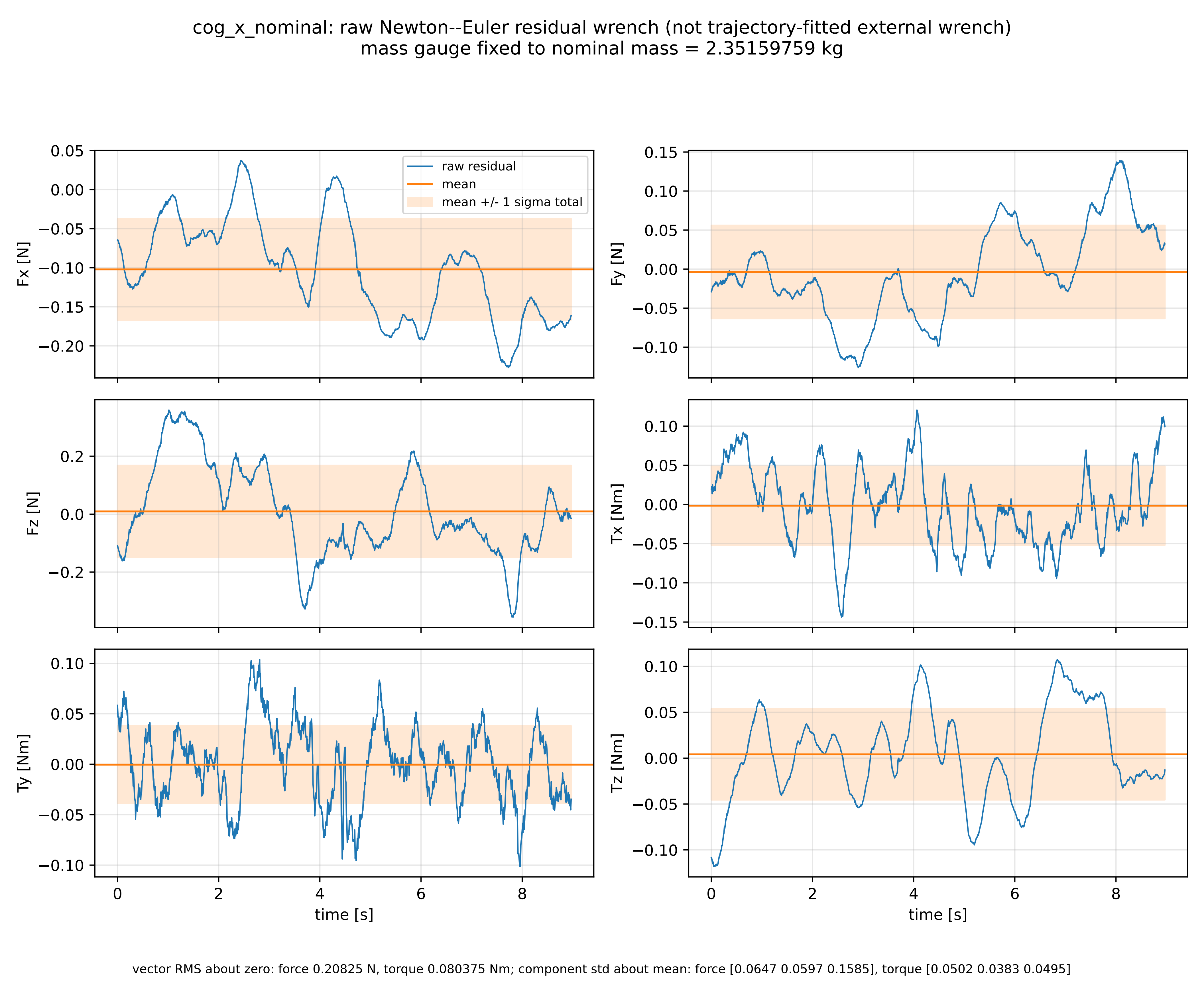
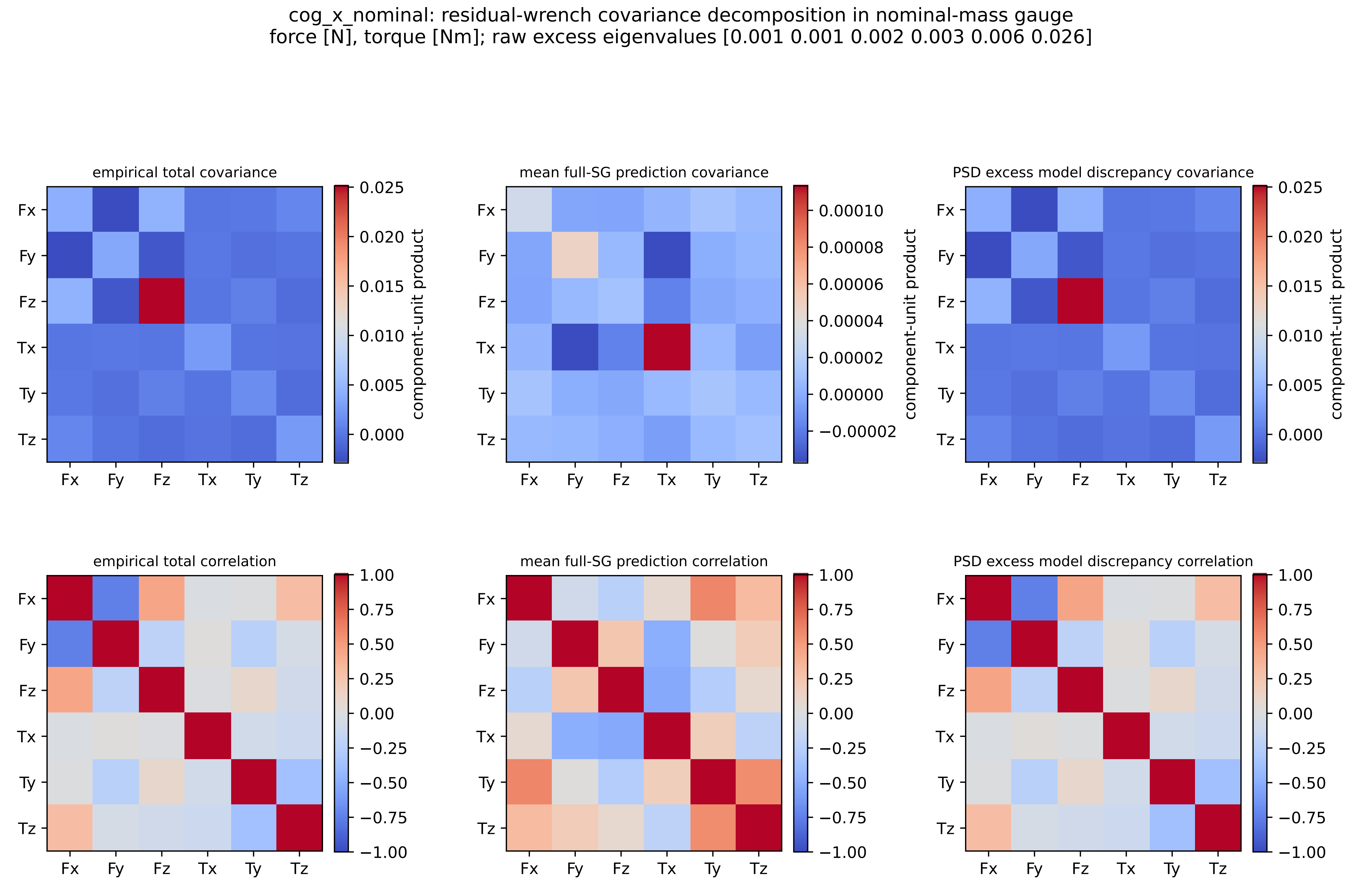
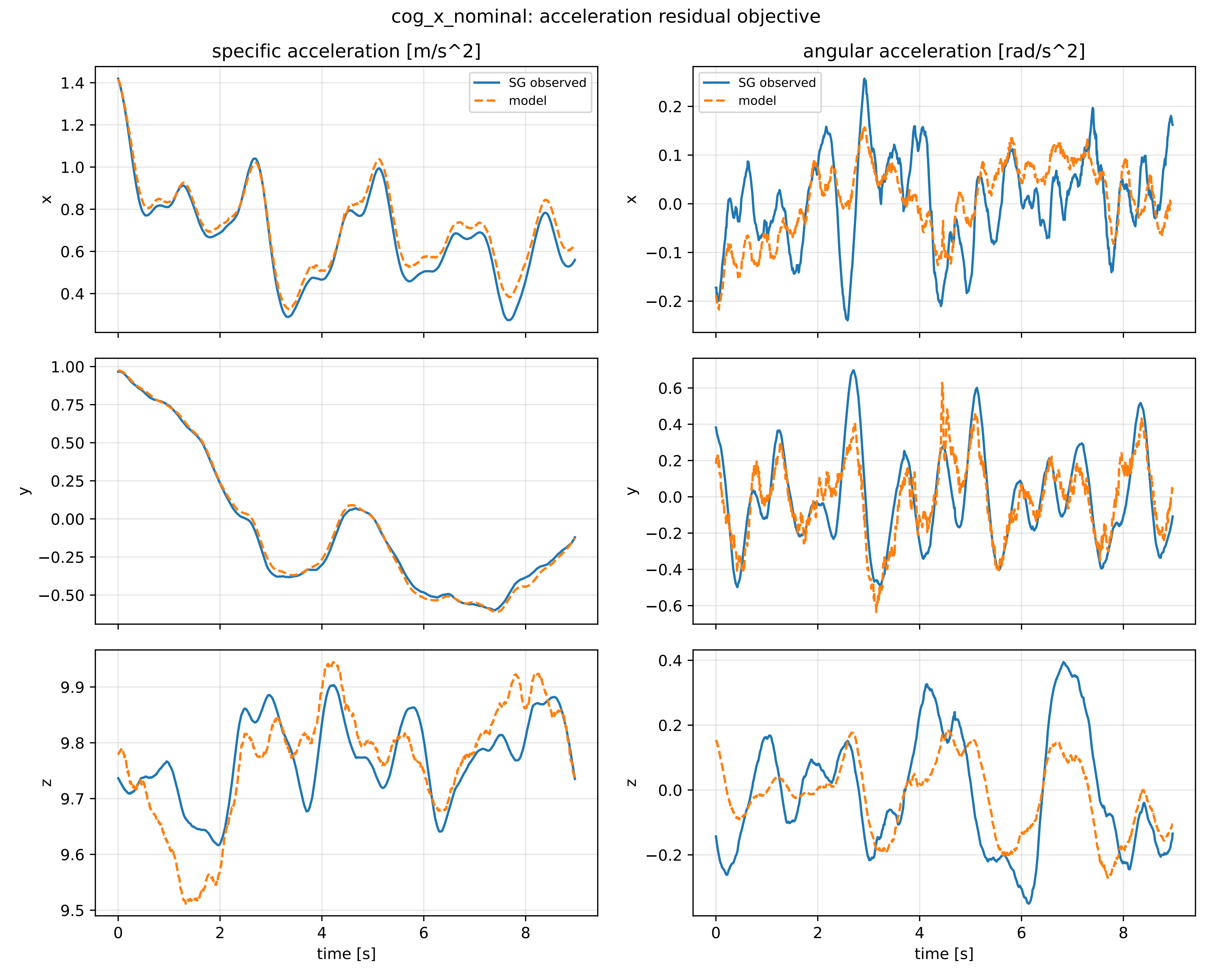
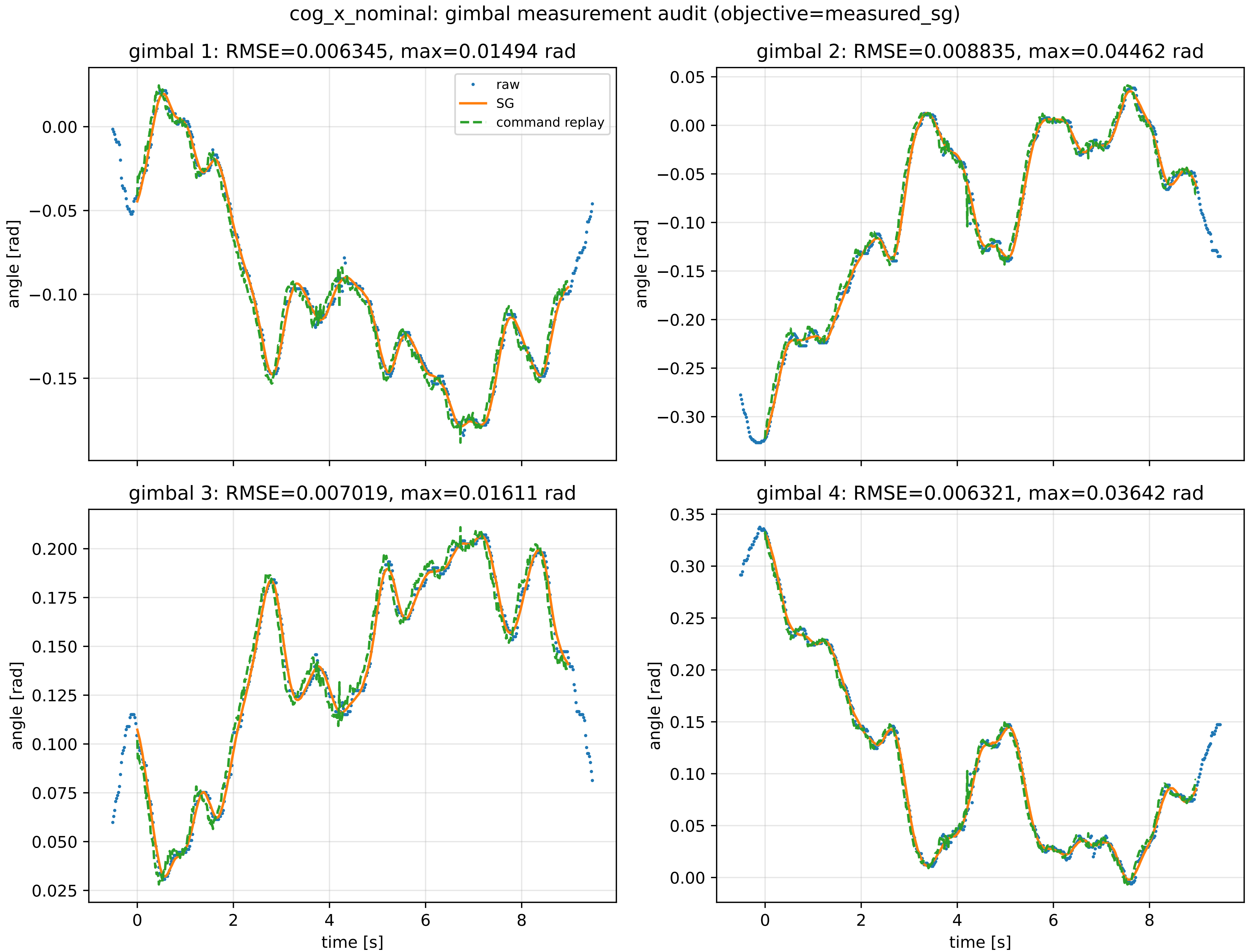

cog_x_nominal: SG trajectory and derived motion
pose position [m]
1.00
0.75
0.50
0.25
0.00
−0.25
x
y
z
body omega [rad/s]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
body alpha [rad/s^2]
0.6
0.4
0.2
0.0
−0.2
−0.4
0
2
4
6
8
time [s]



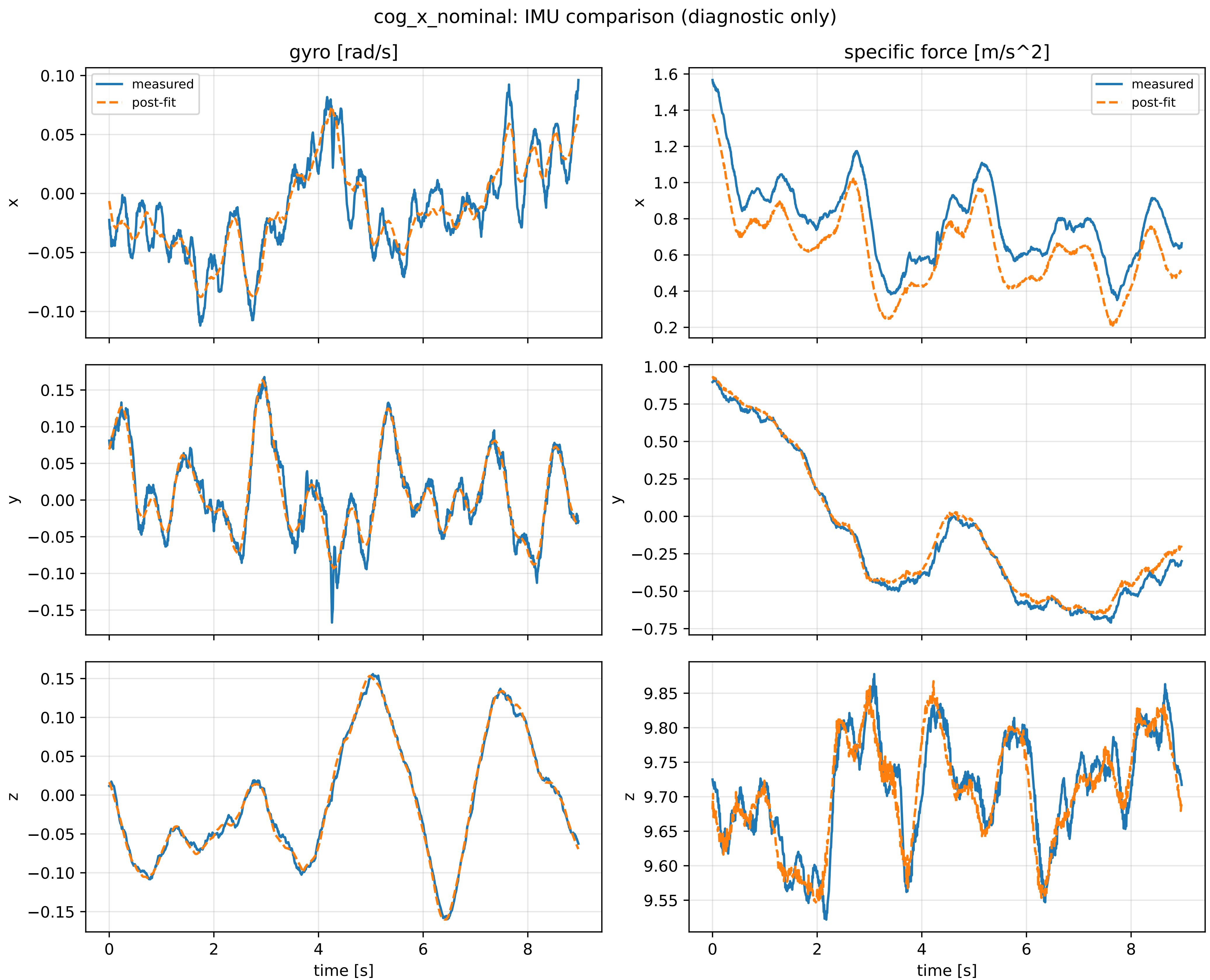
cog_x_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg
raw residual
mean
mean +/- 1 sigma total
Fx [N]
Fy [N]
Fz [N]
Tx [Nm]
Ty [Nm]
Tz [Nm]
time [s]
vector RMS about zero: force 0.20825 N, torque 0.080375 Nm; component std about mean: force [0.0647 0.0597 0.1585], torque [0.0502 0.0383 0.0495]

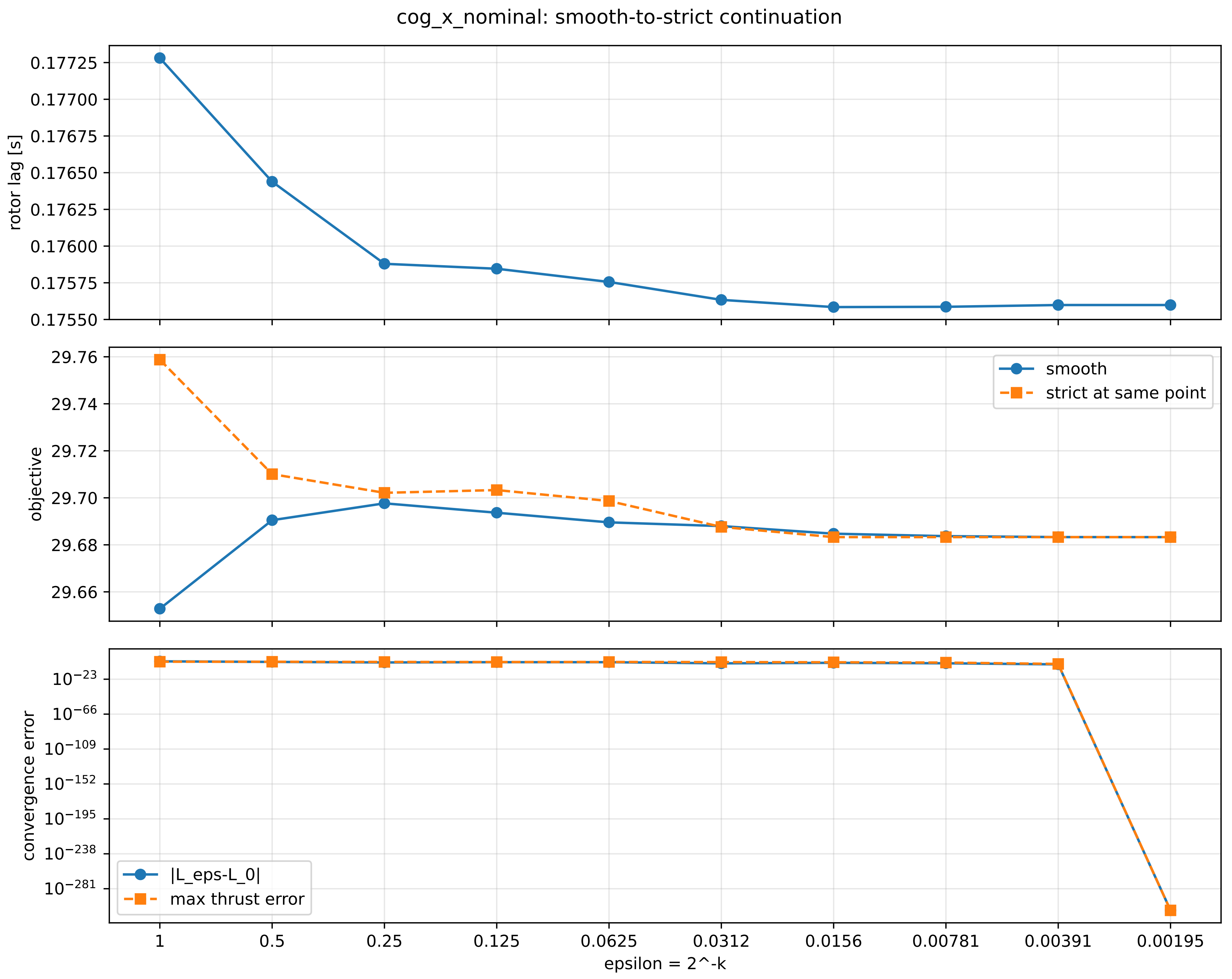

cog_x_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.002 0.003 0.006 0.026]
empirical total covariance
mean full-SG prediction covariance
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
component-unit product
Fx
Fy
Fz
Tx
Ty
Tz
0.025
0.020
0.015
0.010
0.005
0.000
0.00010
0.00008
0.00006
0.00004
0.00002
0.00000
−0.00002
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


cog_x_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



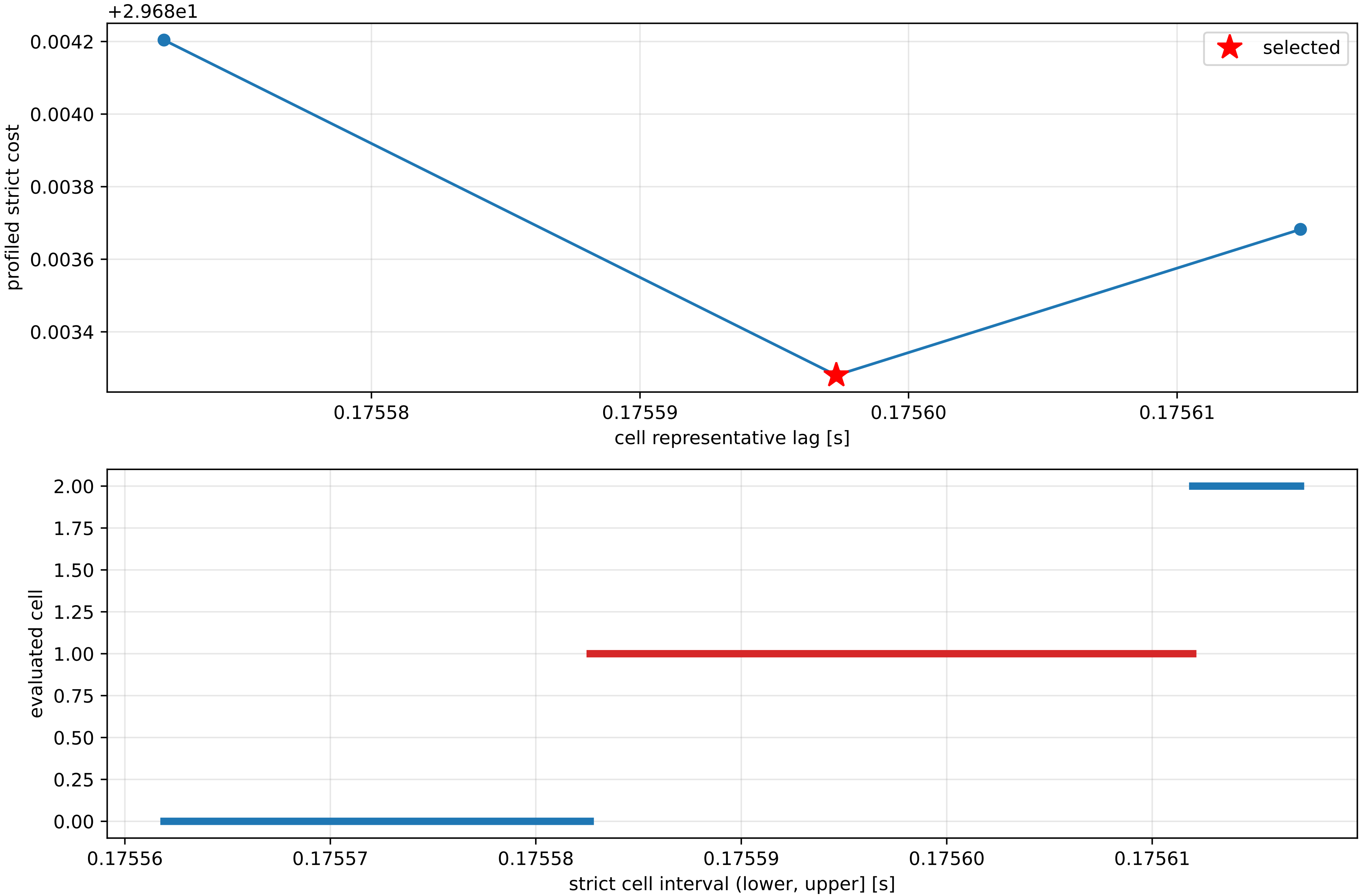
cog_x_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.006345, max=0.01494 rad
gimbal 2: RMSE=0.008835, max=0.04462 rad
gimbal 3: RMSE=0.007019, max=0.01611 rad
gimbal 4: RMSE=0.006321, max=0.03642 rad
raw
SG
command replay
angle [rad]
time [s]


cog_x_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



cog_x_nominal: smooth-to-strict continuation
rotor lag [s]
0.17725
0.17700
0.17675
0.17650
0.17625
0.17600
0.17575
0.17550
objective
29.76
29.74
29.72
29.70
29.68
29.66
smooth
strict at same point
convergence error
10^-23
10^-66
10^-109
10^-152
10^-195
10^-238
10^-281
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

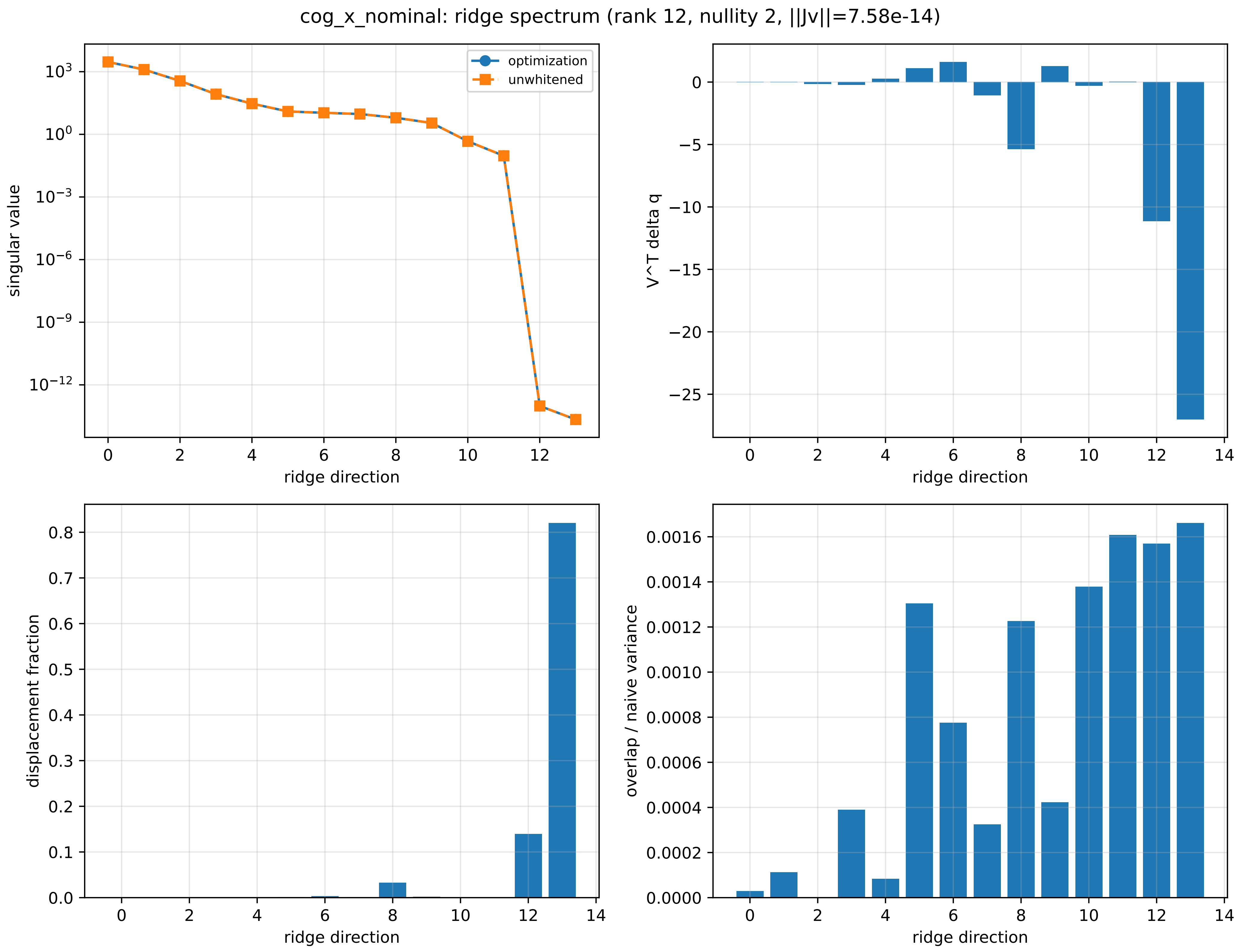
cog_x_nominal: exact strict-ZOH lag cells



selected (0.175582647, 0.175611973] s; final smooth 0.175598577 s; data boundary=False

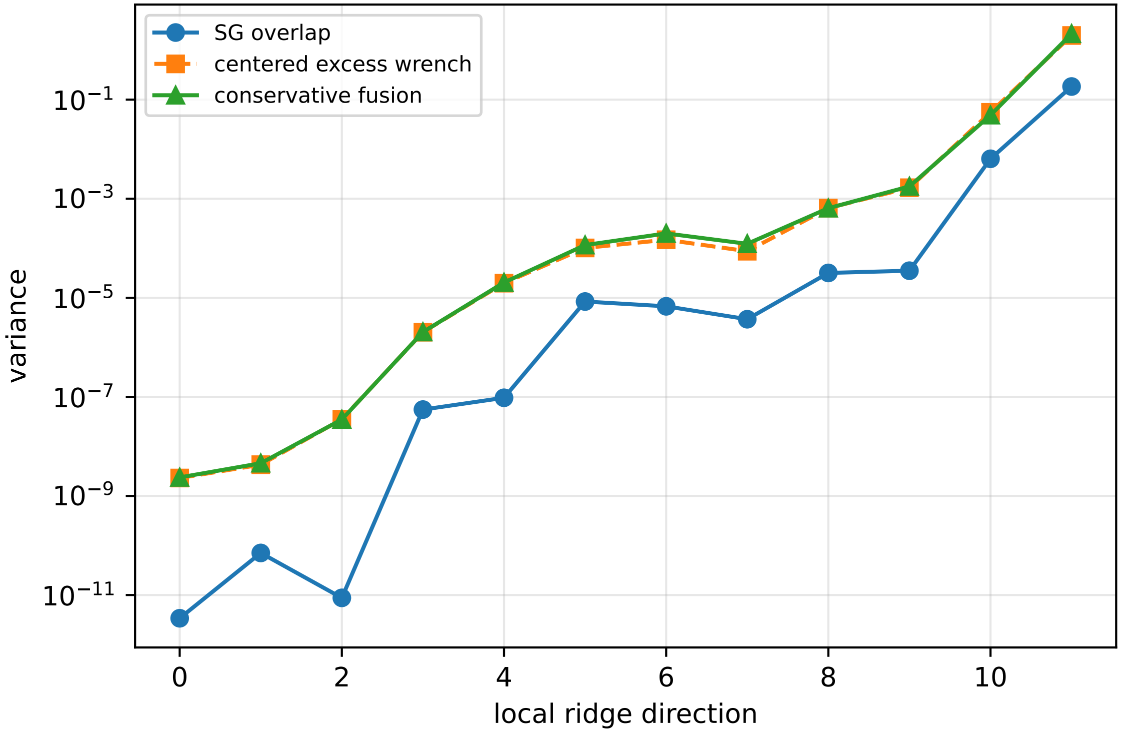
cog_x_nominal: parameter estimate

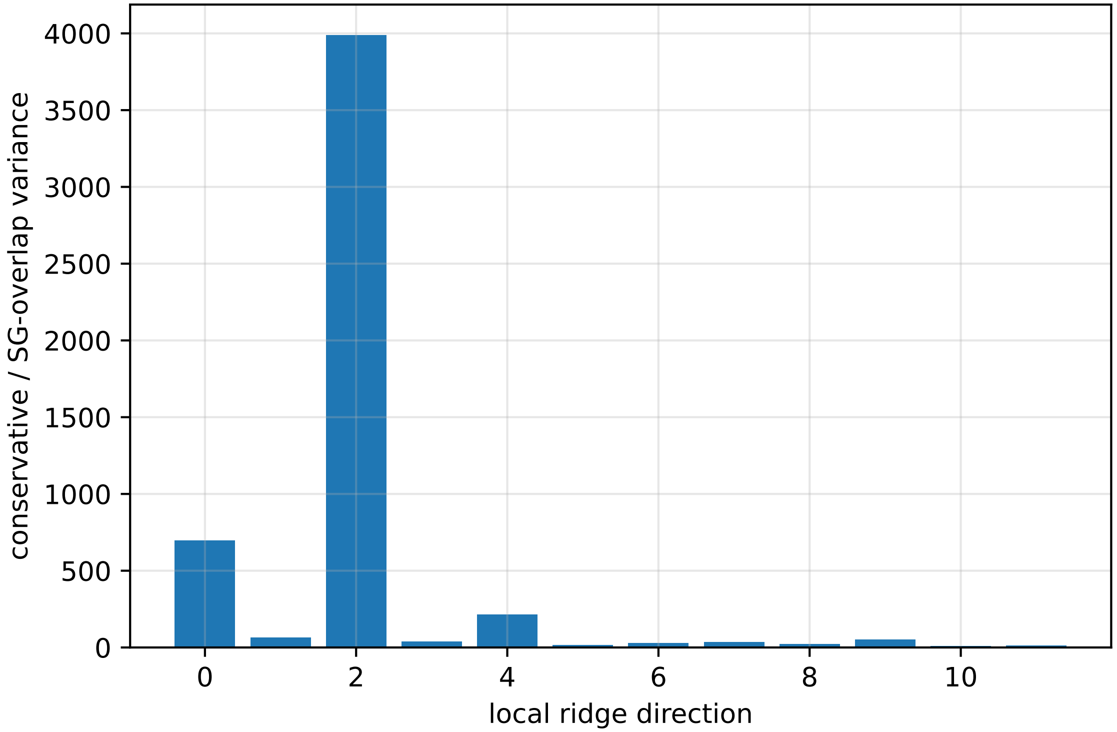
```
Case: cog_x_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 43.088411
inertia [kg m^2]:
[[11.01195753 -0.04492435 -0.64197791]
 [-0.04492435  4.79456532 -0.36467382]
 [-0.64197791 -0.36467382  6.39251551]]
CoG body [m]: [-0.00202472 -0.05857644 -0.00375314]
force effectiveness: [18.09349181 20.38514759 10.74605767 10.82237252]

scale-free J/m [m^2]:
[[ 0.25556657 -0.00104261 -0.01489909]
 [-0.00104261  0.11127273 -0.00846339]
 [-0.01489909 -0.00846339  0.14835812]]
scale-free f/m: [0.4199155  0.47310047 0.24939554 0.25116667]

rotor lag cell: (0.175582647, 0.175611973] s
representative: 0.17559731 s
```

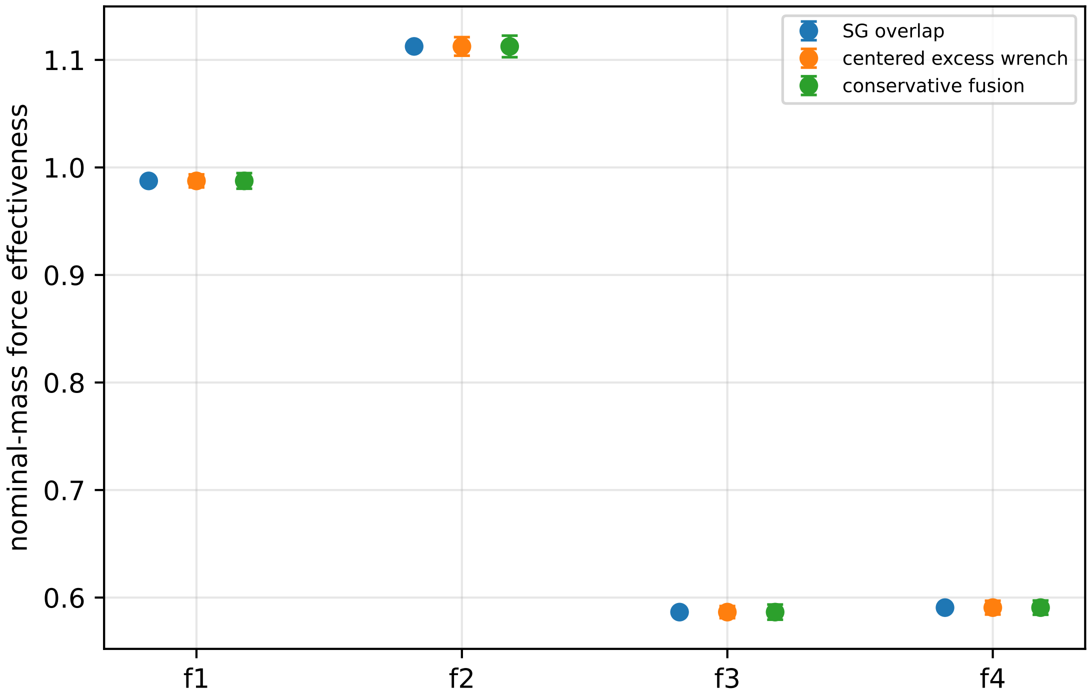


cog_x_nominal: ridge spectrum (rank 12, nullity 2, ||Jv||=7.58e-14)
singular value
optimization
unwhitened
ridge direction
V^T delta q
ridge direction
displacement fraction
ridge direction
overlap / naive variance
ridge direction

cog_x_nominal: Conservative fusion uncertainty

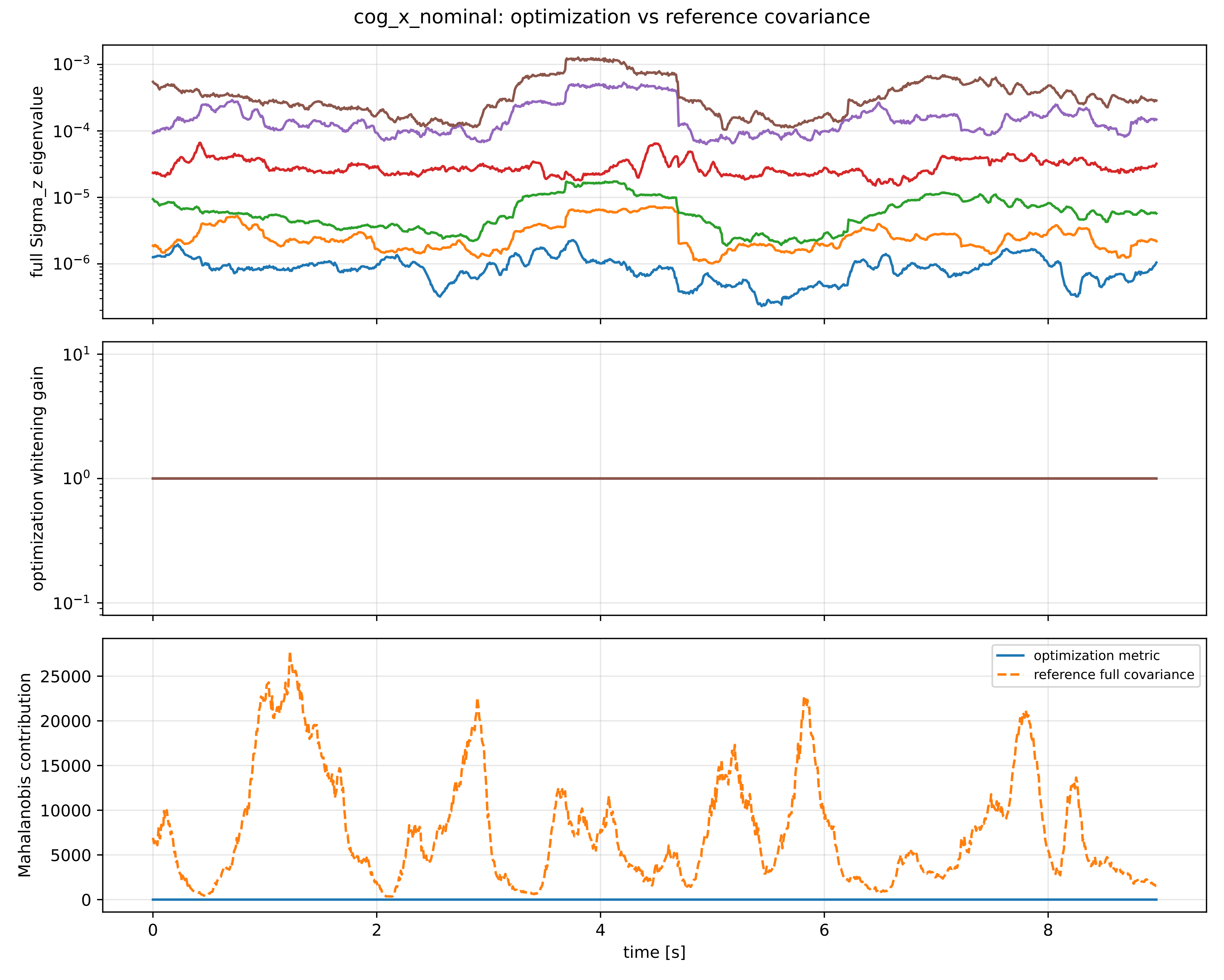


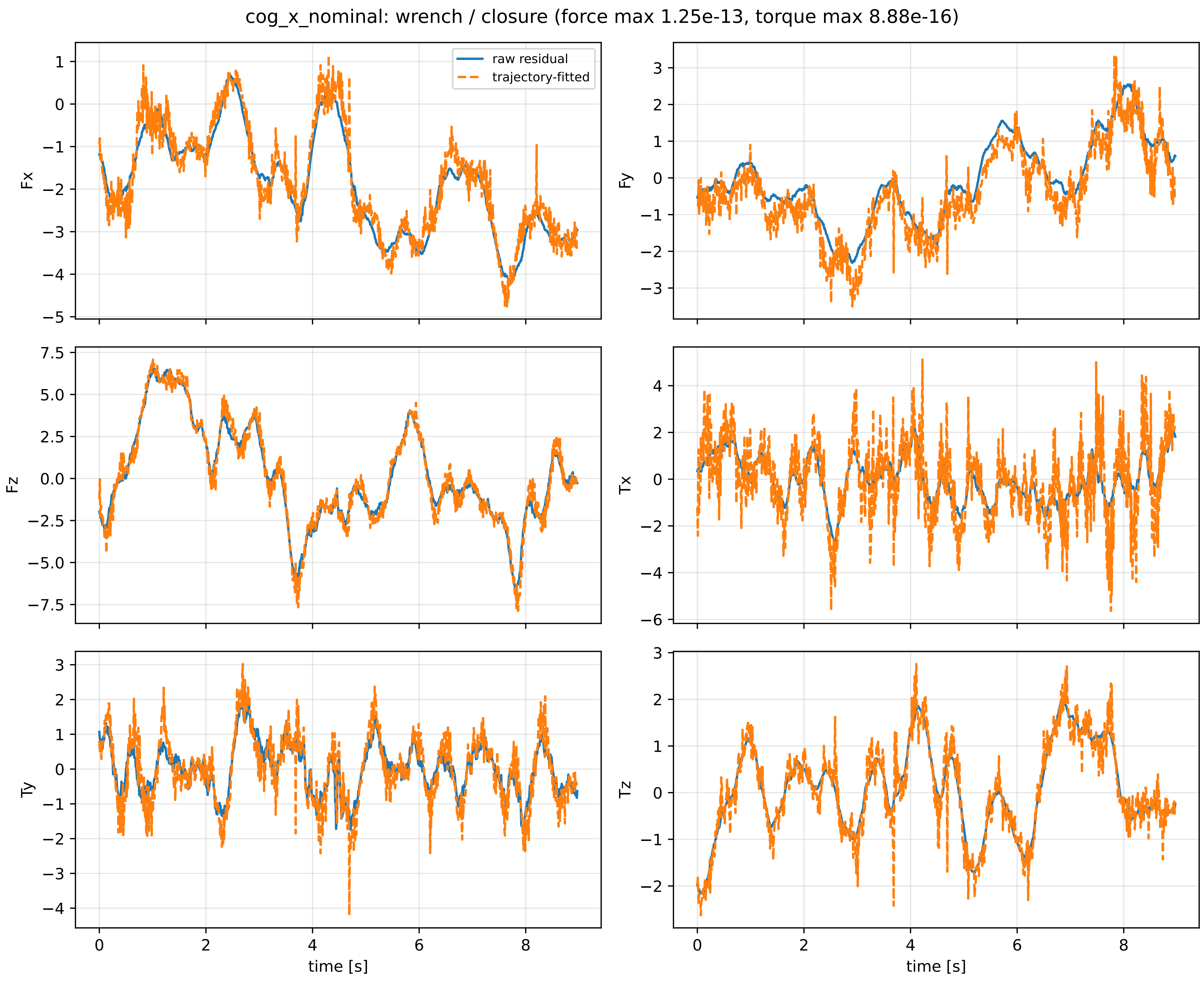





```
Top three non-gauge ambiguous directions
tr(M_res) / tr(M_SG) = 429.8
wrench-to-acceleration recovery max error = 2.9e-15
exact scale gauge ridge index = 12 (alignment 0.924456)

index 11: variance=2.131, ratio=11.59
  log_mass=-0.00123; second_moment_diag_1=-0.00336
  second_moment_diag_2=+0.00184; second_moment_diag_3=+0.00721
  second_moment_offdiag_12=+0.991; second_moment_offdiag_13=-0.134
  second_moment_offdiag_23=-0.0147; cog_x=-0.000343
  cog_y=+0.000114; cog_z=-0.000265
  log_force_eff_1=-3.3e-05; log_force_eff_2=-0.00316
  log_force_eff_3=-0.00184; log_force_eff_4=+0.000569

index 10: variance=0.0486, ratio=7.611
  log_mass=+0.0101; second_moment_diag_1=+0.0167
  second_moment_diag_2=-0.00361; second_moment_diag_3=-0.064
  second_moment_offdiag_12=+0.134; second_moment_offdiag_13=+0.988
  second_moment_offdiag_23=+0.004; cog_x=+0.00321
  cog_y=-0.000296; cog_z=-0.000385
  log_force_eff_1=-0.000754; log_force_eff_2=+0.0224
  log_force_eff_3=+0.0292; log_force_eff_4=-0.01

index 9: variance=0.001768, ratio=50.27
  log_mass=-0.166; second_moment_diag_1=+7.9e-05
  second_moment_diag_2=+0.899; second_moment_diag_3=-0.0636
  second_moment_offdiag_12=+0.00165; second_moment_offdiag_13=+0.00408
  second_moment_offdiag_23=+0.203; cog_x=+0.00889
  cog_y=-0.00178; cog_z=-0.00785
  log_force_eff_1=-0.204; log_force_eff_2=-0.117
  log_force_eff_3=-0.134; log_force_eff_4=-0.214
```

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.


cog_x_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]


cog_x_nominal: wrench / closure (force max 1.25e-13, torque max 8.88e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

cog_x_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_cog_x_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/cog_x_nominal.j
SHA256: bbfefd2485bf27a6773ee0d77fa915418fa80f5a855431ac4e09df6ae9248b8d

data objective:  29.6832793773
prior objective: 8.88647036943e-07
total objective: 29.6832802659

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: cog_x_nominal (cog_position_body_m)
  components: x
  target: [-0.00202471]
  std:    [1.e-05]
  final:  [-0.00202472]
  error:  [-1.33315193e-08]
  standardized residual: [-0.00133315]
  factor objective: 8.88647036943e-07
```